AWS Diagram Lab Report

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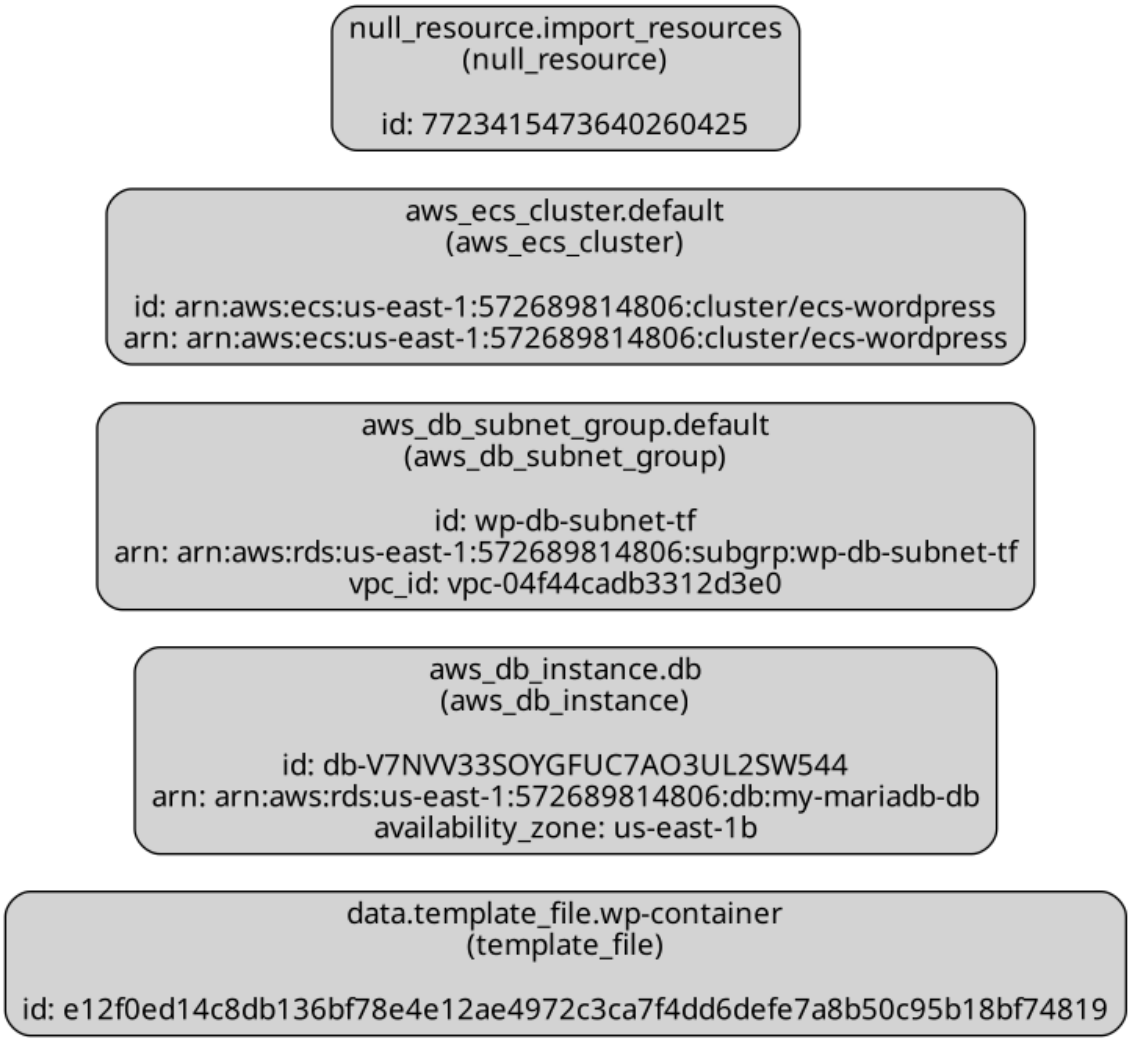
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Architecture Diagram Report

This section describes the AWS architecture created in the lab, including core components such as VPC, subnets, EC2 instances, security groups, RDS, WordPress, ECS cluster, and load balancers. The architecture is fully represented using Terraform and diagrammatically visualized through automated Python scripts.

File contents

a) Terraform Configuration (main.tf)
b) Python Script for Diagram Generation
c) Diagram



d) ARCHITECTURE.md Summary
The documentation outlines architecture goals, design principles, resource relationships, networking layout, security considerations, and operational workflows.

e) Summary-

This lab project focused on automating the end-to-end AWS architecture documentation process. Instead of manually creating diagrams and documentation—which often becomes outdated as infrastructure changes—this solution leverages Terraform for infrastructure-as-code and Python-based diagram generation to ensure accuracy and consistency. The goal was to create a system where infrastructure in Terraform automatically leads to updates in diagrams and architecture files.

To achieve this, existing AWS resources were imported into Terraform using data sources. This allowed the infrastructure to be represented declaratively without modifying production settings. Once resources were defined in Terraform, Python scripts using the diagrams library parsed the Terraform state and generated both technical and management-friendly diagrams. The technical diagram offers detailed visibility into subnets, routing, instances, and monitoring tools, while the management diagram focuses on high-level relationships and business relevance. Importing existing infrastructure such as RDS, WordPress, and AWS WAF can be complex. A Bash script was created to automate the import of these resources using AWS CLI commands. This script is invoked directly from the main.tf file, allowing Terraform to incorporate existing resources seamlessly into its state. Although the instructions reference importing an existing AWS WAF, the current WordPress project does not include any WAF, so no WAF import is performed at this stage. The setup, however, is implemented to accommodate future WAF imports.

Another challenge faced was to create a diagram using Python for the whole infrastructure synchronized with the Terraform configuration. Attached is the “automated architecture diagram based on the imported infrastructure.” I tried all the steps mentioned and imported the image generated from Python, but it was not successful.